

Name:

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CHE 321; Spring - 2020; Quiz - 4

100 Points

A reversible reaction, $A + B \leftrightarrow C + D$ ($\Delta H_{rxn} = -100 \text{ kJ/mol}$), is taking place in a CSTR at adiabatic conditions. The rate law for the reaction is:

$$-r_A = k \left(C_A C_B - \frac{1}{K} C_C C_D \right)$$

$$K_0 = 3.5673; H_1/R = 600.32 \text{ K}$$

$$K = K_0 \exp\left(\frac{H_1}{RT}\right)$$

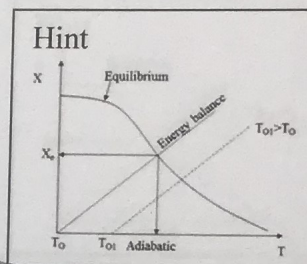
1. If for a particular run in the CSTR, the conversion is found to be 0.8, what can you say about the maximum exit temperature possible for the CSTR? [50]

$$K = \frac{C_{Ce} \cdot C_{De}}{C_{Ae} \cdot C_{Be}} = 16$$

$$\frac{1}{K} = \frac{-r_A}{C_A C_B} = \frac{C_C C_D}{C_A C_B + \frac{r_A}{K}}$$

$$\frac{C_C C_D}{C_A C_B + \frac{r_A}{K}} = 3.5673 \exp\left(\frac{600.32 \text{ K}}{T}\right)$$

$$\ln\left(\frac{C_C C_D}{3.5673(C_A C_B + \frac{r_A}{K})}\right) = \frac{600.32 \text{ K}}{T} \Rightarrow T_{max} = \frac{600.32}{\ln\left(\frac{C_C C_D}{3.5673(C_A C_B + \frac{r_A}{K})}\right)}$$



2. For a CSTR with heat exchanger (typical for the reaction above) in which the above reaction is being carried out, the following equation represents the energy balance. [25]

$$\frac{\dot{Q}}{T} + \underbrace{[-\Delta H_{rxn}^o(T_R) - \Delta \hat{C}_p(T - T_R)] F_{Ao} X}_{2} + \underbrace{(-F_{Ao}) \sum_{i=1}^n \Theta_i \tilde{C}_{pi}(T - T_{io})}_{3} = 0$$

This equation has been divided into three parts. Which of these parts represent heat added to the system and which of these represent heat removed from the system? [25]

Heat Added (positive quantities) = Part 2 $[-\Delta H_{rxn}^o(T_R) - \Delta \hat{C}_p(T - T_R)] F_{Ao} X$ and Part 3 $\dot{Q}$
Heat Removed (negative quantities) = Part 1 $\frac{\dot{Q}}{T}$

3. For the reaction in question #1, the reactants enter the CSTR at 275 K and the CSTR is run adiabatically. If $F_{Ao} = F_{Bo} = 8 \text{ mol/s}$, $C_{PA} = C_{PB} = C_{PC} = C_{PD} = 400 \text{ J/(mol}^\circ\text{C)}$, and final temperature is 375 K, find the value of the heat change due to convection/flow. [25]

$$\dot{Q} + [-\Delta H_{rxn}^o] F_{Ao} X = (F_{Ao} \left[\left(\frac{F_{Ao}}{F_{Ao}} C_{PA} \right) + \left(\frac{F_{Bo}}{F_{Ao}} C_{PB} \right) \right] (275 - 375 \text{ K}))$$

$$(8 \text{ mol/s}) [400 \text{ J/mol}^\circ\text{K} + 400 \text{ J/mol}^\circ\text{K}] \cdot (375 - 275 \text{ K}) = 640 \text{ kJ/s}$$

4. What is the conversion for problem #3?

$$X = \frac{\text{Heat Change}}{F_{Ao} (-\Delta H_{rxn})} = \frac{640 \text{ kJ/s}}{(8 \text{ mol/s}) (-100 \text{ kJ/mol})} = \frac{640 \text{ kJ/s}}{800 \text{ kJ/s}} = 0.80$$